

TEAM ASSIGNMENT #4

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
Professor: Dr. Michael Caracotsios
Office: EIB 244
E-mail: mcaracot@uic.edu

Your name: _____

Due Date: Friday October 18th by 5:00pm in my office

Number of points.....100

Your score....._____

Note: All problems are worth 100 points. Your final score will be the average value

Problem 4A: Refrigeration with Three Stages of Compression

We wish to remove 3MW from a process gas stream in a chiller operating at -35°C and reject it to the environment in a condenser operating at 35°C using propane as the working fluid. Consider the addition of a third stage into the two stage refrigeration cycle you simulated in Assignment #3. Modify the flowsheet accordingly and using the Optimization Flowsheeting option determine the optimal interstage pressures for the three-stage cycle. What is the COP of the cycle for the optimal pressures and how does it compare with the COP of the two-stages compression configuration. Assume that all compressors operate at 75% isentropic efficiency. Start with your own simulation for Assignment #3

Problem 4B: Cyclohexane production

For the cyclohexane production process of **Assignment #3**, reconfigure the flowsheet using unit operations that correspond to actual equipment in the field. For heat exchangers only, if you are going to use any, use two heaters as demonstrated in classroom lectures. Prepare a list of all equipment (except the reactor) and calculate the total capital cost. Prepare a list with all utilities and calculate the total operating cost, including the cost of raw materials, revenues from the product and credits for fuel. Starting with your own simulation from **Assignment #3**. Plant capacity is 300 KMTA of cyclohexane with plant operating data defined in **Assignment #3**. Cooling water is available by a nearby cooling tower that has been designed for dry bulb temperature of 85 deg F, relative humidity of 70% and an approach temperature of 10 deg C. The maximum hot water temperature was taken as 45 deg C for the design of the cooling tower. Multiple pressure levels of steam are available by a steam plant nearby.

REQUIRED REPORT SECTIONS

- Summary of results for both problems (about one page)
- Flowsheets for both processes
- Heat and Weight Balance
- List of Equipment for Problem 4B with the capital costs (see example below)
- List of Utilities/Raw Material/Product for Problem 4B with operating costs

1	2	A	B	C	D	E	F	G
1		Cyclohexane Production Cost Estimate						
2		EQUIPMENT LIST: CAPITAL COSTS						
3			Project:	Assignment #2	Chemical Engineers:			
4			Location:	Chemical Engineering Department	Date:			
5		Equipment Class	ASPEN BlockID	Service Description	Tag	Material Cost	Installed Cost	Annual Cost, \$/yr
6		Distillation						
7		Towers	xxxx	Finishing Column	T-1	\$1,950,958	\$4,095,200	\$819,040
8		Towers	xxxx	Packing, Trays, Internals	T-1			
9		Drums						
10		Vertical Tank	xxxx	Separation Drum	V-1			
11		Vertical Tank	xxxx	Reflux Drum	V-2			
12		Heat Exchangers						
13		S&T Exchanger		Feed-Product Exchanger	E-1			
14		S&T Exchanger		Product Cooler	E-2			
15		Air Coolers						
16		Air Cooler		Reflux Condenser	EA-1			
17		Air Cooler			EA-2			
18		Pumps						
19		Pump		Reflux Pump	P-1			
20		Pump			P-2			
21		Turbines						
24		Compressors						
27		Reactors						
30								